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University of Kentucky Researchers Develop “Bring Your Own Projector” Machine Vision Camera

A single board pairs a conventional industrial camera with a built-in HDMI controller, turning any projector into a structured-light 3D scanner — and chaining camera to camera over the video signal itself.

LEXINGTON, Ky. — Researchers at the University of Kentucky have built a machine vision camera that does something conventional industrial cameras do not: it speaks HDMI.

The board is a traditional high-speed machine vision camera — a global-shutter sensor streaming full frames over USB 3.0 — with a complete HDMI receiver and transmitter integrated alongside it on the same FPGA. Video from a host computer enters the camera and passes through to a projector. Because the camera sits *in* the video path rather than beside it, it knows precisely what is being projected, and when.

The team calls the result BYOP, for *Bring Your Own Projector*: connect any HDMI projector and the system becomes a structured-light 3D scanner, with no external trigger box, frame grabber, or custom synchronization cabling. The reason to want that is economic as much as technical. Structured-light systems are conventionally built around dedicated projection light engines — metrology modules made in small volumes, at prices that dominate the cost of the instrument. Commodity projectors are mass-produced against the 3D-Ready and DLP market instead, and they win on nearly every axis that matters to a scanner: far greater luminance, a wide choice of fields of view and throw distances, and, with LED illumination, an operating life measured in tens of thousands of hours. What kept them out of precision scanning was never their optics — it was the difficulty of making a camera keep time with a device never designed to be part of a measuring instrument.

Why put HDMI inside a camera?

Structured-light 3D scanning works by projecting a sequence of patterns onto an object and photographing each one. The measurement is only as good as the synchronization: the camera must expose once per projected pattern, at a known moment within that pattern’s display time.

Conventionally, that means a trigger generator, a cable harness, and a calibration step for the delay between them — an assembly that has to be reworked whenever the projector, the frame rate, or the pattern set changes.

Integrating the video interface into the camera removes the problem rather than managing it. The

projected frames and the camera's exposures come from the same signal, inside one device.

The imager

The camera is built around the **onsemi PYTHON 1300**, a global-shutter CMOS sensor with 1280×1024 resolution — 1.3 megapixels — reading out over four LVDS channels at 720 Mbit/s each. The sensor is socketed, and the same board accepts either the **monochrome or the colour** variant of the part, so the choice is made per application rather than per design.

Pixels are delivered at **10 bits** of depth, not the 8 bits typical of consumer imaging: 1024 grey levels instead of 256. For structured light, that matters directly — the depth resolution of a phase measurement follows from how finely the camera can distinguish one projected intensity from the next, so the extra two bits are measurement precision, not just headroom.

At full resolution and 10-bit depth, the sensor is capable of **210 frames per second** — comfortably more than a USB link can carry continuously.

For live streaming, the camera delivers **120 frames per second** over its USB-C connection, at the sensor's full 1280×1024 and full 10-bit depth — no cropping, no windowing, and no reduction in depth to buy the rate. Pixels are packed 10-to-a-word rather than padded into 16 bits, which is what makes a full-resolution frame fit the link at that rate.

For scans that need to be faster still, the camera does not have to stream at all. Frames are captured into **on-board DDR3 memory** at the sensor's own rate and downloaded afterwards, so the length of a scan is set by the sensor rather than by the host connection.

A global shutter means every pixel in a region is exposed at the same instant, so nothing shears when the subject or the camera is moving.

No host HDMI source required

The camera does not have to be fed video at all. The board generates its own pattern sequences internally and drives the projector directly, at frame rates up to **120 per second** — the same rate at which it streams full frames.

That rate is not arbitrary. **3D-Ready DLP projectors** — a large and inexpensive installed base, built to accept 120 Hz for shutter-glasses stereo — take a 120 Hz signal as a matter of course. The camera drives them at their native rate, and the highest standard mode it drives at 120 Hz is **800×600**. Anything above that is a 60 to 75 Hz proposition.

The entire structured-light loop — project a pattern, capture it, advance to the next — therefore runs on the camera itself, at 120 Hz, with nothing driving the video path but the camera. A computer is needed only to collect the resulting images, not to generate what is projected.

Where a host *is* driving the display, the camera passes its video through untouched, so the same hardware serves both arrangements.

Any projector — and a host offered only what works

Nothing about the camera is tied to a particular projector. On power-up, it reads the projector's **EDID** — the small data block every display serves back over the video cable to describe what it can accept — and works from what it finds there. In standalone operation, it drives the projector at a mode the projector actually advertises, rather than at an assumed one.

When a host PC is in the video path, the camera goes a step further: it builds an EDID of its own and serves that to the computer. What the PC is offered is the projector's mode list *narrowed to the range the camera can itself receive* — from **640×480 at 60 Hz** at the bottom to **1280×800 at 60 Hz** at the top, taking in 1024×768 at up to 75 Hz along the way. The camera is working to a pixel-rate budget rather than a single maximum, so resolution and frame rate trade against one another: the smaller the frame, the faster it may arrive.

Because many projectors serve a placeholder EDID while they warm up, the camera keeps re-reading it and re-offers the host an updated list whenever the content changes — so the projector's real modes appear without anyone unplugging a cable.

Daisy-chaining: one video signal, many cameras

Because the HDMI path runs *through* the camera, cameras can be chained. The host drives the first unit; each unit forwards the video onward to the next and, finally, to the projector. Every camera in the chain sees the same frames and can key to them.

That makes multi-camera geometries straightforward to assemble:

- **Active stereo** — two or more cameras viewing a projector-illuminated scene, capturing the same pattern from different viewpoints.
- **Multi-view capture** — cameras arranged around a subject, all keyed to a common projected sequence.
- **Extended baselines** — units placed far apart, coordinated over the same cable that already carries the video.

The hardware

Image sensor	onsemi PYTHON 1300, socketed — monochrome or colour
Resolution	1280 × 1024 (1.3 MP), global shutter
Pixel depth	10 bits per pixel (1024 grey levels); 8-bit selectable
Sensor frame rate	210 frames/s at full resolution, 10-bit
Streamed rate	120 frames/s sustained over USB-C
Buffered capture	To on-board DDR3 at sensor rate, downloaded after
Host interface	USB 3.0, 325 MB/s sustained, byte-exact
Video in / out	HDMI receive and transmit, pass-through
Projector support	Reads the projector's EDID; any projector in the window
Pass-through window	25–90 MHz pixel clock, enforced in the EDID served to the host
Standalone output	Board-generated video, no HDMI source needed
Highest 120 Hz mode	800×600 at 120 Hz (3D-Ready DLP projectors)
Verified modes	640 × 480 through 1280 × 800, 60 to 120 Hz
Processing	Xilinx Artix-7 FPGA with DDR3 frame buffer
Construction	Alchitry Pt V2, Ft+ and Hd+ boards plus a custom sensor board
On-board patterns	Sinusoidal fringe generation, radiometric linearization

The FPGA generates structured-light fringe patterns on the board itself and applies a host-uploadable correction curve, so that a commanded intensity ramp reproduces linearly on the projector — a calibration that otherwise has to be carried out in host software for every projector.

The instrument is built as a stack. Three of its boards are commercially available parts from **Alchitry** — the Pt V2 mainboard carrying the Artix-7 FPGA, the Ft+ for the USB 3.0 link, and the Hd+ for HDMI in and out — and on top of them sits a **custom-designed sensor board**,

developed for this project, carrying the PYTHON 1300 and its power supplies. The boards mate through a common high-density stacking connector, so the whole camera assembles without a single internal cable and drops into the printed enclosure shown below.

Background — the work this camera comes from

The architecture announced here was proposed and argued for in this group’s earlier published work. What is new is that it has been built, and measured. Readers who want the reasoning behind the design will find it in the following.

- M. P. Ruffner, Y. Yu and D. L. Lau, “Structured light smart camera for spatial augmented reality applications,” *Proc. SPIE* **10932**, Emerging Digital Micromirror Device Based Systems and Applications XI, 109320J (2019). [doi:10.1117/12.2508553](https://doi.org/10.1117/12.2508553)
The direct predecessor. It sets out why a PC cannot be trusted to synchronize a projector and a camera, and proposes putting the image sensor and an HDMI input and output onto the FPGA controller itself — the design this camera realizes.
- M. P. Ruffner, “Design of a Machine Vision Camera for Spatial Augmented Reality,” MSEE thesis, University of Kentucky (2018). uknowledge.uky.edu/ece_etds/129
The long-form treatment of the same design, including the power and layout work that a sensor board of this kind stands or falls on.
- Y. Yu, D. L. Lau, M. P. Ruffner and K. Liu, “Dual-projector structured light 3D shape measurement,” *Applied Optics* **59**(4), 964–974 (2020). [doi:10.1364/AO.378363](https://doi.org/10.1364/AO.378363)
Why more than one light engine is worth the trouble: fewer occlusions, better signal-to-noise, fewer frames per scan, and multi-path made visible. The argument that makes a chainable camera worth building.
- Y. Yu and D. L. Lau, “3D scanning by means of dual-projector structured light illumination,” *Proc. SPIE* **10932**, 109320M (2019). [doi:10.1117/12.2510990](https://doi.org/10.1117/12.2510990)
The conference treatment of the above, with the temporal-frequency multiplexing that lets two projectors run at once.

“This system is the culmination of years of research and development into structured light imaging, and by making commodity projector synchronization so easy, I hope it unlocks years more of technological improvements in SLI.”

— Daniel L. Lau, DataBeam Professor of Electrical and Computer Engineering, University of Kentucky

Images

Additional images and captions available on request.

About

The BYOP camera comes out of the laboratory of **Daniel L. Lau**, DataBeam Professor of Electrical and Computer Engineering at the University of Kentucky, whose career in three-dimensional imaging spans twenty-five years. Since joining the University of Kentucky in 2001, he has worked on structured-light 3D imaging — both the algorithms that make real-time, high-accuracy shape measurement possible and the instruments that have to carry them — work that established non-contact 3D fingerprint capture as a practical alternative to contact scanning and that has reached



The complete camera, opened. The board stack sits in its printed enclosure: the HDMI board at the bottom, the USB 3.0 image link and the Artix-7 FPGA above it, and the PYTHON 1300 sensor board on top, its glass lid facing up. The 48-pin sensor, on a 1.016 mm lead pitch, was soldered by hand in the lab. At right, the enclosure's lid carries the lens and drops straight onto the stack. *Photo: University of Kentucky.*

industry through two companies he co-founded, FlashScan3D and Seikowave. He is also a pioneer of digital halftoning, and in 2026 was elected a Fellow of the IEEE for contributions to digital printing and 3D imaging.

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The connector edge. The two HDMI ports at the bottom are the video input and output. The *input* is fed by the host PC, or by the previous camera when units are daisy-chained; the *output* drives the projector, or the next camera in the chain. Above them are two USB-C connectors: the lower carries the image stream to the host, the upper is the FPGA board's own configuration and diagnostic link. The sensor board is the topmost in the stack. *Photo: University of Kentucky.*